🧠 YPD CVP LITE™ - PROGRESSIVE SCORING LOGIC SYSTEM

📊 INJECTABLE SCORING FRAMEWORK FOR VENDOR IMPLEMENTATION

🔧 Core Scoring Architecture (Injectable Code)

```
class CVPLiteScoringEngine:
    def __init__(self):
        self.assessment_weights = self.load_step_weights()
        self.scoring_hidden = True  # No visibility until Step 10
        self.progressive_scores = {}
        self.interconnection_matrix = self.build_interconnection_map()

    def load_step_weights(self):
        """
        Define which steps are scored and their psychological significance
        """
        return {
            0: {"scored": False, "type": "profile_setup", "weight": 0},
            1: {"scored": True, "type": "interest_discovery", "weight": 15},
            2: {"scored": True, "type": "values_assessment", "weight": 15},
            3: {"scored": True, "type": "personality_learning", "weight": 20},
            4: {"scored": True, "type": "passion_alignment", "weight": 20},
            5: {"scored": True, "type": "cognitive_scenarios", "weight": 15},
            6: {"scored": True, "type": "emotional_intelligence", "weight": 15},
            7: {"scored": False, "type": "career_synthesis", "weight": 0},  # Synthesis step
            8: {"scored": False, "type": "education_guidance", "weight": 0},  # Guidance step
            9: {"scored": False, "type": "action_planning", "weight": 0},  # Planning step
            10: {"scored": False, "type": "report_generation", "weight": 0}  # Final compilation
        }

    def build_interconnection_map(self):
        """
        Define how each step's insights connect to build final career profile
        """
        return {
            "career_clarity": {
```

```
            "primary_contributors": [1, 4, 7],  # Interest + Passion + Synthesis
            "supporting_contributors": [2, 3],   # Values + Personality
            "weight_distribution": [40, 40, 20]
        },
        "self_awareness": {
            "primary_contributors": [2, 3, 6],  # Values + Personality + EI
            "supporting_contributors": [1, 4],   # Interest + Passion
            "weight_distribution": [35, 35, 30]
        },
        "decision_making_maturity": {
            "primary_contributors": [5, 6, 9],  # Cognitive + EI + Planning
            "supporting_contributors": [2, 3],   # Values + Personality
            "weight_distribution": [40, 30, 30]
        },
        "growth_potential": {
            "primary_contributors": [3, 5, 6],  # Personality + Cognitive + EI
            "supporting_contributors": [1, 2, 4], # Interest + Values + Passion
            "weight_distribution": [40, 35, 25]
        }
    }
```

🎯 STEP-SPECIFIC SCORING ALGORITHMS (Injectable)

📈 Step 1: Interest & Strengths Discovery Scoring

```
def score_interest_discovery(self, mcq_responses, journal_reflection):
    """
    Score based on interest breadth, depth, and self-awareness
    """
    scoring_components = {
        "interest_breadth": self.analyze_interest_diversity(mcq_responses),
        "interest_depth": self.assess_passion_intensity(mcq_responses),
        "self_reflection_quality": self.evaluate_journal_depth(journal_reflection),
        "authenticity_indicators": self.detect_genuine_responses(mcq_responses)
    }
```

```
    # Progressive scoring - builds foundation for later steps
    interest_score = {
        "raw_score": self.calculate_weighted_score(scoring_components, [25, 25, 30, 20]),
        "interest_profile": self.generate_interest_taxonomy(mcq_responses),
        "reflection_depth": scoring_components["self_reflection_quality"],
        "authenticity_level": scoring_components["authenticity_indicators"]
    }

    # Store for interconnected scoring in later steps
    self.progressive_scores[1] = interest_score
    return interest_score

def analyze_interest_diversity(self, responses):
    """
    Measure breadth of interests across different domains
    """
    interest_domains = {
        "analytical": 0, "creative": 0, "social": 0, "practical": 0,
        "investigative": 0, "entrepreneurial": 0
    }

    for response in responses:
        domain = self.map_response_to_domain(response)
        interest_domains[domain] += 1

    diversity_score = len([domain for domain, count in interest_domains.items() if count > 0])
    return min(diversity_score / 4.0, 1.0) * 100  # Normalized to 100

def evaluate_journal_depth(self, journal_text):
    """
    Assess quality of self-reflection based on depth and authenticity
    """
    depth_indicators = {
        "word_count": len(journal_text.split()),
```

```
        "emotional_words": self.count_emotional_language(journal_text),
        "future_connections": self.detect_future_thinking(journal_text),
        "specific_examples": self.count_concrete_examples(journal_text),
        "question_exploration": self.detect_questioning_mindset(journal_text)
    }

    # Weighted scoring for reflection quality
    depth_score = (
        min(depth_indicators["word_count"] / 50, 1.0) * 20 +  # Adequate length
        depth_indicators["emotional_words"] * 15 +            # Emotional awareness
        depth_indicators["future_connections"] * 25 +         # Forward thinking
        depth_indicators["specific_examples"] * 20 +          # Concrete thinking
        depth_indicators["question_exploration"] * 20         # Curious mindset
    )

    return min(depth_score, 100)
```

📊 **Step 2: Values & Work Preferences Scoring**

```
def score_values_assessment(self, mcq_responses, journal_reflection):
    """
    Score based on value clarity, consistency, and work-life alignment
    """
    # Interconnect with Step 1 for consistency checking
    interest_foundation = self.progressive_scores.get(1, {})

    scoring_components = {
        "value_clarity": self.assess_value_definition(mcq_responses),
        "work_life_integration": self.evaluate_work_preferences(mcq_responses),
        "consistency_with_interests": self.check_interest_value_alignment(
            mcq_responses, interest_foundation
        ),
        "reflection_maturity": self.evaluate_values_reflection(journal_reflection),
        "decision_framework": self.assess_decision_making_values(mcq_responses)
    }
```

```
    values_score = {
        "raw_score": self.calculate_weighted_score(scoring_components, [20, 20, 25, 20, 15]),
        "core_values_profile": self.generate_values_taxonomy(mcq_responses),
        "work_environment_match": scoring_components["work_life_integration"],
        "consistency_index": scoring_components["consistency_with_interests"],
        "decision_maturity": scoring_components["decision_framework"]
    }

    # Update interconnected scoring
    self.progressive_scores[2] = values_score
    return values_score

def check_interest_value_alignment(self, current_responses, previous_interest_data):
    """
    Measure consistency between stated interests and underlying values
    """
    if not previous_interest_data:
        return 75  # Default score if no previous data

    interest_profile = previous_interest_data.get("interest_profile", {})
    current_values = self.extract_value_indicators(current_responses)

    alignment_score = 0
    total_comparisons = 0

    for interest_type, interest_strength in interest_profile.items():
        expected_values = self.get_expected_values_for_interest(interest_type)
        actual_values = current_values

        for expected_value in expected_values:
            if expected_value in actual_values:
                alignment_score += interest_strength * actual_values[expected_value]
                total_comparisons += interest_strength
```

```
    return (alignment_score / total_comparisons * 100) if total_comparisons > 0 else 50
```

🧠 **Step 3: Learning Style + Personality Assessment Scoring**

```
def score_personality_learning(self, mcq_responses, journal_reflection):
    """
    Score based on self-awareness, learning style clarity, and personality insights
    """
    # Interconnect with previous steps for comprehensive profile building
    previous_data = {
        "interests": self.progressive_scores.get(1, {}),
        "values": self.progressive_scores.get(2, {})
    }

    scoring_components = {
        "learning_style_clarity": self.assess_learning_preferences(mcq_responses),
        "personality_self_awareness": self.evaluate_personality_insights(mcq_responses),
        "behavioral_consistency": self.check_behavioral_patterns(mcq_responses, previous_data),
        "adaptation_potential": self.assess_flexibility_indicators(mcq_responses),
        "metacognitive_awareness": self.evaluate_learning_reflection(journal_reflection)
    }

    personality_score = {
        "raw_score": self.calculate_weighted_score(scoring_components, [20, 25, 20, 15, 20]),
        "learning_style_profile": self.generate_learning_taxonomy(mcq_responses),
        "personality_dimensions": self.extract_personality_factors(mcq_responses),
        "behavioral_consistency": scoring_components["behavioral_consistency"],
        "growth_mindset_indicators": scoring_components["adaptation_potential"]
    }

    self.progressive_scores[3] = personality_score
    return personality_score
```

💔 **Step 4: Passion-Career Alignment Scoring**

```
def score_passion_alignment(self, mcq_responses, journal_reflection):
    """
    Score based on passion depth, career connection, and realistic goal-setting
```

```
    """
    # This is a KEY interconnection step - synthesizes Steps 1-3
    comprehensive_profile = {
        "interests": self.progressive_scores.get(1, {}),
        "values": self.progressive_scores.get(2, {}),
        "personality": self.progressive_scores.get(3, {})
    }

    scoring_components = {
        "passion_authenticity": self.assess_genuine_passion(mcq_responses, journal_reflection),
        "career_connection_clarity": self.evaluate_career_links(mcq_responses),
        "goal_realism": self.assess_realistic_expectations(mcq_responses, comprehensive_profile),
        "motivation_depth": self.analyze_intrinsic_motivation(journal_reflection),
        "integration_success": self.measure_profile_synthesis(mcq_responses, comprehensive_profile)
    }

    passion_score = {
        "raw_score": self.calculate_weighted_score(scoring_components, [25, 20, 20, 20, 15]),
        "passion_profile": self.generate_passion_taxonomy(mcq_responses),
        "career_alignment_index": scoring_components["career_connection_clarity"],
        "motivation_sustainability": scoring_components["motivation_depth"],
        "profile_integration": scoring_components["integration_success"]
    }

    self.progressive_scores[4] = passion_score
    return passion_score
```

🧩 **Step 5: Cognitive Problem-Solving Scenarios Scoring**

```
def score_cognitive_scenarios(self, scenario_responses, journal_reflection):
    """
    Score based on reasoning quality, problem-solving approach, and decision-making maturity
    """
    scoring_components = {
        "logical_reasoning": self.assess_reasoning_quality(scenario_responses),
        "creative_problem_solving": self.evaluate_solution_creativity(scenario_responses),
```

```
        "decision_making_process": self.analyze_decision_approach(scenario_responses),
        "stakeholder_consideration": self.assess_perspective_taking(scenario_responses),
        "reflection_on_thinking": self.evaluate_metacognitive_journal(journal_reflection)
    }

    cognitive_score = {
        "raw_score": self.calculate_weighted_score(scoring_components, [25, 20, 25, 15, 15]),
        "reasoning_style_profile": self.generate_cognitive_taxonomy(scenario_responses),
        "problem_solving_strengths": scoring_components["creative_problem_solving"],
        "decision_maturity": scoring_components["decision_making_process"],
        "social_cognition": scoring_components["stakeholder_consideration"]
    }

    self.progressive_scores[5] = cognitive_score
    return cognitive_score

def assess_reasoning_quality(self, responses):
    """
    Evaluate logical thinking and systematic problem-solving approach
    """
    reasoning_indicators = {
        "systematic_approach": 0,
        "evidence_consideration": 0,
        "cause_effect_thinking": 0,
        "alternative_evaluation": 0
    }

    for response in responses:
        # Analyze choice patterns for logical consistency
        if self.shows_systematic_thinking(response):
            reasoning_indicators["systematic_approach"] += 1
        if self.considers_evidence(response):
            reasoning_indicators["evidence_consideration"] += 1
        if self.shows_causal_thinking(response):
```

```
            reasoning_indicators["cause_effect_thinking"] += 1
        if self.evaluates_alternatives(response):
            reasoning_indicators["alternative_evaluation"] += 1

    total_indicators = sum(reasoning_indicators.values())
    max_possible = len(responses) * 4  # 4 indicators per response

    return (total_indicators / max_possible * 100) if max_possible > 0 else 0
```

❤️ **Step 6: Emotional Intelligence & Ethics Scoring**

```
def score_emotional_intelligence(self, scenario_responses, journal_reflection):
    """
    Score based on emotional awareness, empathy, ethical reasoning, and social intelligence
    """
    scoring_components = {
        "emotional_self_awareness": self.assess_emotion_recognition(scenario_responses),
        "empathy_and_perspective": self.evaluate_empathy_indicators(scenario_responses),
        "ethical_reasoning": self.analyze_moral_decision_making(scenario_responses),
        "social_intelligence": self.assess_relationship_awareness(scenario_responses),
        "emotional_regulation": self.evaluate_emotion_management(journal_reflection)
    }

    ei_score = {
        "raw_score": self.calculate_weighted_score(scoring_components, [20, 25, 25, 15, 15]),
        "emotional_profile": self.generate_ei_taxonomy(scenario_responses),
        "empathy_level": scoring_components["empathy_and_perspective"],
        "ethical_maturity": scoring_components["ethical_reasoning"],
        "social_skills_potential": scoring_components["social_intelligence"]
    }

    self.progressive_scores[6] = ei_score
    return ei_score
```

🔗 **INTERCONNECTED FINAL SCORING SYNTHESIS (Injectable)**

🎯 **Step 10: Final Report Score Calculation**

```
def generate_final_career_profile_scores(self):
    """
    Synthesize all progressive scores into comprehensive career readiness profile
    This is the ONLY place where scores become visible to students
    """
    if not self.all_scored_steps_completed():
        raise Exception("Cannot generate final scores - assessment incomplete")

    # Calculate interconnected domain scores
    final_profile = {
        "career_clarity_score": self.calculate_career_clarity(),
        "self_awareness_score": self.calculate_self_awareness(),
        "decision_maturity_score": self.calculate_decision_maturity(),
        "growth_potential_score": self.calculate_growth_potential(),
        "overall_readiness_score": 0  # Calculated after domain scores
    }

    # Calculate overall readiness as weighted average of domain scores
    final_profile["overall_readiness_score"] = self.calculate_overall_readiness(final_profile)

    # Generate qualitative insights based on score patterns
    final_profile["qualitative_insights"] = self.generate_narrative_insights(final_profile)

    # Create action-oriented recommendations
    final_profile["development_recommendations"] = self.create_development_plan(final_profile)

    return final_profile

def calculate_career_clarity(self):
    """
    Synthesize interests, passion alignment, and career synthesis for clarity score
    """
    interconnection = self.interconnection_matrix["career_clarity"]
    contributors = interconnection["primary_contributors"]
```

```
    weights = interconnection["weight_distribution"]

    # Get scores from contributing steps
    interest_data = self.progressive_scores[1]
    passion_data = self.progressive_scores[4]
    # Step 7 synthesis is qualitative, use passion alignment as proxy

    clarity_components = {
        "interest_definition": interest_data.get("authenticity_level", 0),
        "passion_career_connection": passion_data.get("career_alignment_index", 0),
        "integration_success": passion_data.get("profile_integration", 0)
    }

    # Weighted calculation
    clarity_score = (
        clarity_components["interest_definition"] * (weights[0]/100) +
        clarity_components["passion_career_connection"] * (weights[1]/100) +
        clarity_components["integration_success"] * (weights[2]/100)
    )

    return {
        "score": round(clarity_score, 1),
        "level": self.categorize_score_level(clarity_score),
        "key_strengths": self.identify_clarity_strengths(clarity_components),
        "growth_areas": self.identify_clarity_growth_areas(clarity_components)
    }

def calculate_self_awareness(self):
    """
    Synthesize values, personality, and emotional intelligence for self-awareness score
    """
    values_data = self.progressive_scores[2]
    personality_data = self.progressive_scores[3]
    ei_data = self.progressive_scores[6]
```

```
    awareness_components = {
        "values_clarity": values_data.get("decision_maturity", 0),
        "personality_insight": personality_data.get("behavioral_consistency", 0),
        "emotional_awareness": ei_data.get("empathy_level", 0)
    }

    weights = self.interconnection_matrix["self_awareness"]["weight_distribution"]

    awareness_score = (
        awareness_components["values_clarity"] * (weights[0]/100) +
        awareness_components["personality_insight"] * (weights[1]/100) +
        awareness_components["emotional_awareness"] * (weights[2]/100)
    )

    return {
        "score": round(awareness_score, 1),
        "level": self.categorize_score_level(awareness_score),
        "depth_indicators": self.analyze_awareness_depth(awareness_components),
        "authenticity_markers": self.identify_authenticity_signs(awareness_components)
    }

def categorize_score_level(self, score):
    """
    Convert numerical scores to meaningful descriptive levels
    """
    if score >= 85:
        return "Exceptional - Clear direction and strong self-understanding"
    elif score >= 70:
        return "Strong - Good foundation with some areas for growth"
    elif score >= 55:
        return "Developing - Emerging awareness with growth potential"
    elif score >= 40:
        return "Exploring - Early stages of self-discovery"
```

```
    else:
        return "Beginning - Great starting point for career exploration"
```

📊 SCORE VISIBILITY & REPORTING SYSTEM (Injectable)

🔒 Hidden Scoring During Journey

```
class ScoreVisibilityController:
    def __init__(self):
        self.scores_hidden = True
        self.step_feedback_mode = "qualitative_only"

    def provide_step_feedback(self, step_number, assessment_results):
        """
        Provide encouraging feedback WITHOUT revealing numerical scores
        """
        if self.scores_hidden and step_number < 10:
            return {
                "feedback_type": "qualitative_growth_focused",
                "insights": self.generate_growth_insights(assessment_results),
                "encouragement": self.create_encouraging_message(assessment_results),
                "next_step_preview": self.create_anticipation_builder(step_number + 1),
                "scores": None  # Explicitly hidden
            }
        elif step_number == 10:
            return self.generate_final_comprehensive_report(assessment_results)

    def generate_growth_insights(self, results):
        """
        Create meaningful insights without numerical scoring
        """
        insights = []

        # Focus on discovered strengths
        if results.get("authenticity_level", 0) > 70:
```

```
        insights.append("Your responses show genuine self-reflection - this authenticity will serve you well in any career path.")

    if results.get("consistency_index", 0) > 65:
        insights.append("There's a beautiful alignment between your interests and values, suggesting you have a clear sense of what matters to you.")

    if results.get("growth_mindset_indicators", 0) > 60:
        insights.append("Your openness to learning and growth shines through - this adaptability is a superpower in today's world.")

    return insights

def create_encouraging_message(self, results):
    """
    Build confidence while maintaining growth mindset
    """
    encouragement_bank = [
        "You're building incredible self-awareness through this journey.",
        "Your thoughtful responses show you're taking this seriously - that commitment will pay off.",
        "Every insight you're gaining brings you closer to your ideal career path.",
        "Your authentic reflection is exactly what makes this assessment valuable for you."
    ]

    # Select encouraging message based on engagement level
    engagement_level = self.assess_engagement_quality(results)
    return self.select_appropriate_encouragement(encouragement_bank, engagement_level)
```

🎯 Final Report Score Presentation

```
def generate_comprehensive_final_report(self, all_assessment_data):
    """
    Create the complete career profile report with all scores revealed
    """
    final_scores = self.generate_final_career_profile_scores()

    report_structure = {
```

```
        "student_celebration": self.create_celebration_header(),
        "career_readiness_overview": {
            "overall_score": final_scores["overall_readiness_score"],
            "score_explanation": "This reflects your current career exploration readiness based on self-awareness, interests, values, and decision-making maturity.",
            "level_description": final_scores["overall_readiness_score"]["level"]
        },
        "detailed_profile_scores": {
            "career_clarity": final_scores["career_clarity_score"],
            "self_awareness": final_scores["self_awareness_score"],
            "decision_maturity": final_scores["decision_maturity_score"],
            "growth_potential": final_scores["growth_potential_score"]
        },
        "narrative_insights": final_scores["qualitative_insights"],
        "development_action_plan": final_scores["development_recommendations"],
        "next_steps_guidance": self.create_next_steps_framework(),
        "parent_discussion_guide": self.generate_family_conversation_starters()
    }

    return self.format_final_report(report_structure)

def create_celebration_header(self):
    """
    Celebrate completion and validate the journey
    """
    return {
        "celebration_message": "🎉 Congratulations! You've completed your Career Vision Journey!",
        "journey_validation": "Over the past 10 steps, you've engaged in deep self-reflection, explored your interests and values, and gained valuable insights about yourself.",
        "growth_acknowledgment": "This level of self-exploration shows maturity and commitment to your future - qualities that will serve you well in any path you choose."
    }
```

🔧 **VENDOR IMPLEMENTATION GUIDE (Injectable)**

📝 **Integration Checklist**

```
# Required database schema for scoring system
SCORING_TABLES = {
    "student_progressive_scores": {
        "student_id": "VARCHAR(50) PRIMARY KEY",
        "step_1_scores": "JSON",
        "step_2_scores": "JSON",
        "step_3_scores": "JSON",
        "step_4_scores": "JSON",
        "step_5_scores": "JSON",
        "step_6_scores": "JSON",
        "interconnection_data": "JSON",
        "final_profile_scores": "JSON",
        "scoring_timestamp": "TIMESTAMP",
        "scores_revealed": "BOOLEAN DEFAULT FALSE"
    },
    "assessment_quality_metrics": {
        "student_id": "VARCHAR(50)",
        "step_number": "INT",
        "engagement_quality": "FLOAT",
        "reflection_depth": "FLOAT",
        "authenticity_indicators": "JSON",
        "consistency_tracking": "JSON"
    }
}

# API endpoints for scoring system
SCORING_ENDPOINTS = {
    "POST /api/cvp-lite/score-assessment": "Score individual step assessment",
    "GET /api/cvp-lite/check-scoring-readiness": "Verify if final scoring can proceed",
    "POST /api/cvp-lite/generate-final-profile": "Generate comprehensive final report",
    "GET /api/cvp-lite/get-step-feedback": "Retrieve qualitative feedback only"
}

# Configuration for scoring system
```

```
SCORING_CONFIG = {
    "score_visibility": False,  # Hidden until Step 10
    "minimum_reflection_words": 20,
    "authenticity_threshold": 60,
    "consistency_tolerance": 15,
    "quality_weight_factors": {
        "reflection_depth": 0.3,
        "response_consistency": 0.25,
        "authenticity_indicators": 0.25,
        "engagement_quality": 0.2
    }
}
```

⚙️ **Implementation Priority Order**

1. **Step Assessment Scoring** - Implement individual step scoring algorithms
2. **Progressive Score Storage** - Set up secure score accumulation system
3. **Interconnection Logic** - Build score synthesis and relationship mapping
4. **Quality Validation** - Implement authenticity and consistency checking
5. **Final Report Generator** - Create comprehensive profile compilation
6. **Visibility Controls** - Enforce score hiding until final report
7. **Testing Framework** - Validate scoring accuracy and consistency

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🎯 **SCORING SYSTEM READY**: This comprehensive scoring logic provides psychologically sound, interconnected assessment that builds student self-awareness while maintaining engagement through hidden scoring until the transformative final report reveal. The system is production-ready for immediate vendor implementation.**